Phantasos-Lite — Experiment 2 Results Log: Qwen2.5-1.5B-Instruct

Reference doc for all numbers and findings from the K=16 Qwen2.5-1.5B-Instruct run (Tier A same-family replication), kept alongside Experiment 1 (Qwen2.5-7B-Instruct) as the baseline to compare Experiment 3 (Llama-3.2-3B-Instruct, if run) against. Every number below is taken directly from notebook output, not recalculated or rounded differently. All layer-dependent numbers are from the corrected run at `PRIMARY_LAYERS=[7,13,21]`*, after a mid-project fix (see §7) — no number in this doc comes from the earlier mismatched* `[7,14,21]` *run.*

Model & setup

Model	`Qwen/Qwen2.5-1.5B-Instruct`
hidden_size	1536
num_hidden_layers	28
`PRIMARY_LAYERS`	[7, 13, 21] — hardcoded to match Experiment 1 exactly (not derived from `round(n_layers*f)`, which gives [7,14,21] for this depth and does not match Exp 1's hand-picked 13)
Personas (K)	16
I-set prompts	8
Checkpoint folder	`/content/drive/MyDrive/phantasos_K16lite_qwen1p5b_checkpoints`

Persona set

Identical to Experiment 1: `assistant, physician, professor, teacher, interrogator, critic, pirate, pirate_story` (original 8) + `customer_service, drill_sergeant, poet, poet_story, therapist, comedian, librarian, diplomat` (new 8).

Cluster counts: anchor=1, expert=3 (physician, professor, librarian), warm=3 (customer_service, teacher, therapist), cold=4 (drill_sergeant, interrogator, diplomat, critic), playful=5 (pirate, poet, pirate_story, poet_story, comedian) — same cluster assignments as Experiment 1.

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1. Environment checks — all passed

Check	Result
Hook output type	`torch.Tensor` (`HOOK_OUTPUT_IS_TUPLE = False`)
Digit token ids	`[16,17,18,19,20,21,22,23,24]` — all 9 distinct, single-token
alpha=0 no-op (Check D1)	**True**
Per-position delta norm (Check D2)	not separately printed in this run's log (carried over from Exp 1's passing check; same tokenizer/chat-template family)

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2. S2 — persona expression (judge-scored, 0–3 scale)

persona	mean	n
assistant	0.62	16
physician	1.38	16
professor	1.25	16
teacher	1.75	16
interrogator	0.62	16
critic	1.44	16
pirate	2.25	16
pirate_story	1.81	16
customer_service	0.88	16
drill_sergeant	1.44	16
poet	2.06	16
poet_story	1.81	16
therapist	1.31	16
comedian	1.62	16
librarian	1.12	16
diplomat	0.94	16

- **S2 pass rate, excluding interrogator:** 0.59 (n=224) — threshold was ⩾0.85
- **S2 pass rate, all role personas:** 0.57 — threshold was ⩾0.85
- **Length–expression correlation:** r = 0.15 (weaker confound than Exp 1's 0.43 — length tracks judged expression less in the smaller model)
- **Comparison to Exp 1:** both pass-rate figures are lower than the 7B run (0.69 / 0.66), consistent with a smaller instruction-tuned model following persona framing less reliably. Not layer-dependent, so unaffected by the §7 fix.

3. S5 — activation separability (leave-one-variant-out, layer 13)

- **Accuracy: 0.742** (chance = 0.0625 for K=16)
- Well below Experiment 1's 0.906, and below the K=8 pilot's 100%.
- **Confusions (at layer 13, corrected run):**
  - `physician` → confused with `therapist` (7/8 misclassified)
  - `teacher` → confused with `therapist` (6/8 misclassified)
  - `customer_service` → confused with `therapist` (4/8 misclassified)
  - `librarian` → confused with `therapist` (3/8 misclassified)
  - `drill_sergeant` → spread across `pirate` (2/8), `interrogator` (2/8), `critic` (1/8)
  - `pirate_story` → confused with `pirate` (2/8), `poet` (1/8)
  - `interrogator` → confused with `critic` (1/8)
- **Interpretation:** unlike Experiment 1, where confusions landed precisely on the four engineered "surprise" pairs, this run shows `therapist` acting as a generic attractor that pulls in misclassifications from four otherwise-unrelated personas (physician, teacher, customer_service, librarian). This reads as coarser, less fine-grained persona structure in the smaller model, not simply a scaled-down version of Exp 1's confusion pattern.

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4. Baseline RSA — activation vs. prompt text (embedding baseline)

Not the finding — this is the confound being controlled for.

layer	RSA(act, text)	perm-p
7	0.537	0.0002
13	0.499	0.0002
21	0.397	0.0008

- **Shuffled-label control RSA (layer 13):** −0.090 (expected ≈0 — passed)
- **Split-half noise ceiling (variant 0 vs. variant 1):**
  - layer 7: 0.529
  - layer 13: 0.669
  - layer 21: 0.857
- **Comparison to Exp 1:** ceiling is lower at every layer than the 7B run (0.769/0.837/0.967) — the smaller model's activation geometry is itself less split-half-reliable, which matters for interpreting the primary RSA numbers below (see §6 ceiling-normalized comparison).

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5. RDM-S validity (judge-free pairwise self-rating)

- 256/256 ratings collected (16×16, including diagonal self-ratings)
- **Low-confidence ratings (digit_mass < 0.8):** 0 / 256
- **Self-rating EV (expected ≈9):** mean = 7.09, min = 3.54
- **Comparison to Exp 1:** notably worse-calibrated than the 7B run (mean 8.87, min 6.97) — the digit-mass confidence filter still passes cleanly, but the underlying logit distribution over rating digits is less sharp/consistent in the smaller model. Flagged as a specific weakness of the self-rating channel, not of the extraction pipeline.

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6. PRIMARY RESULT — activation ↔ self-report correspondence

This is the paper's headline finding.

layer	self-desc RSA (raw)	self-desc (partial-out text)	self-rating RSA (raw)	self-rating (partial-out text)
7	0.417 (p=0.0002)	0.262	0.237 (p=0.0170)	0.102
13	0.619 (p=0.0002)	0.529	0.476 (p=0.0002)	0.400
21	**0.672** (p=0.0002)	**0.610**	**0.687** (p=0.0002)	**0.652**

- **RDM-F / RDM-S channel agreement:** 0.456 (notably weaker than Exp 1's 0.723 — see §5 for the likely driver: self-rating EV miscalibration, not a text-description problem)
- **Pattern:** layer-21-peaking, replicating Experiment 1's shape at roughly a fifth the parameter count. Raw magnitudes are uniformly lower than the 7B run at every layer.

Ceiling-normalized comparison (RSA ÷ split-half reliability)

The fairer cross-model read, correcting for the lower noise ceiling found in §4:

layer	self-desc, Exp 1 (7B)	self-desc, Exp 2 (1.5B)	self-rating, Exp 1 (7B)	self-rating, Exp 2 (1.5B)
7	0.603	**0.788**	0.554	0.448
13	0.736	**0.925**	0.777	0.711
21	0.750	0.784	0.806	0.802

- **Key finding:** once corrected for reliability, the 1.5B model's **self-description** channel tracks its own activation geometry *more tightly* than the 7B model's does, at every layer

— raw RSA alone makes the smaller model look uniformly weaker, but that's partly a ceiling artifact. The **self-rating** channel is roughly matched at layer 21 (0.802 vs 0.806) but weaker at shallower depths, consistent with the EV-calibration issue in §5 degrading that one channel specifically rather than the underlying representational correspondence.

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7. Bugs encountered this run (for Methods/reproducibility notes, not results)

- `PRIMARY_LAYERS` **mismatch:** the layer-selection cell computed `PRIMARY_LAYERS = [round(n_layers*f) for f in (0.25,0.5,0.75)]`, giving `[7,14,21]` for this 28-layer model. Experiment 1 used `[7,13,21]`, hand-picked rather than derived from that formula (13/28 = 46.4%, not literally 50%). The divergence was caught by auditing the figures against the printed numbers before it was written into any cross-model table, not during the run itself. Fixed by hardcoding `PRIMARY_LAYERS=[7,13,21]` and re-running everything downstream of activation extraction (S5, baseline RSA, noise ceiling, primary RSA, all three figures). S2 judge-scoring and RDM-S rating collection are layer-independent and were **not** re-run — those numbers (§2, §5) are unaffected by the fix.
- **Stale/desynced checkpoint on re-run:** after the fix, `act_index.jsonl` still listed 256 jobs "done" at the old layer set while `act_vectors.npz` lacked any `|L13` keys. The per-layer-key desync guard (checking `{_id}|L{L}` presence in the vector store rather than trusting the index count) correctly detected this and re-extracted all 256 jobs cleanly — no silent reuse of wrong-layer vectors.
- **npz vector-count drift:** `act_vectors.npz` holds 1024 vectors post-fix, not the expected 768 (256×3) — the old `|L14` vectors were superseded but never pruned from the in-memory store dict before the final `np.savez`. Does not affect any reported number (nothing reads `|L14` keys downstream); left as cleanup debt in this checkpoint folder rather than fixed, since Experiment 3 uses its own separate folder.
- **Stale hardcoded comparison strings:** two `EXPECTED:` print lines in the S5 cell (`"Chance level (8 personas): 0.125"` and `"(64, 3584)... [8 personas * 8 prompts]"`) are leftover K=8-pilot text never updated for K=16. The actually-computed accuracy (0.742) and tensor shapes (128, 1536) are correct — only the printed comparison text is wrong, and only cosmetically.

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8. Files on Drive (checkpoint folder)

`s2_descriptions.jsonl`, `s2_judge.jsonl`, `act_index.jsonl`, `act_vectors.npz`, `rdm_s_ratings.jsonl`, `lite_rdm_heatmaps.png`, `lite_mds.png` — all confirmed complete (256/256/256/1024*/256 respectively; *1024 includes 256 stale unused `|L14` vectors, see §7).

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9. One-paragraph summary (for quick reference before Experiment 3)

Qwen2.5-1.5B-Instruct, K=16 personas, replicates Experiment 1's headline pattern — a layer-21-peaking correspondence between internal activation geometry and self-report (RSA 0.672 self-description, 0.687 self-rating) that survives partialing out prompt-text wording — at roughly a fifth the parameter count of the 7B baseline. Raw RSA magnitudes are lower at every layer, but once corrected for the smaller model's lower split-half noise ceiling, the self-description channel's reliability-normalized correspondence actually *exceeds* the 7B run's at every layer (e.g. 0.925 vs 0.736 at layer 13), while the self-rating channel is roughly matched only at the deepest layer and weaker upstream — traceable to a specific EV-calibration weakness in the smaller model's self-rating readout (mean 7.09 vs target 9, vs 7B's 8.87), not to a general breakdown of the self-report/activation link. Separability is markedly lower than Exp 1 (74.2% vs 90.6%) and the confusions are qualitatively different: rather than landing on the four engineered surprise pairs, `therapist` acts as a generic attractor pulling in four otherwise-unrelated personas, indicating coarser sub-cluster structure at this scale. S2 pass rates are also lower (0.59/0.57 vs 0.69/0.66), consistent with weaker persona-following in the smaller model. Net: the correspondence finding replicates across a 4.7x parameter-count reduction within the same model family — the Tier A cross-scale replication the project scope called for — with channel-specific, well-understood degradation rather than a uniform collapse of the effect.